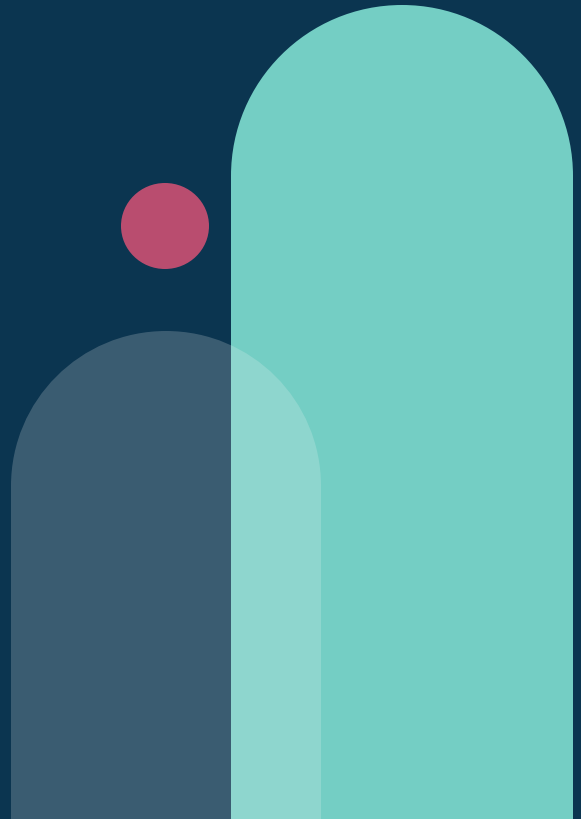
Basics of AI, ML, and DL

Deep Dive into Neural Networks and ML Algorithms

01

Introduction to Neural Networks



Introduction to Neural Networks

Definition:

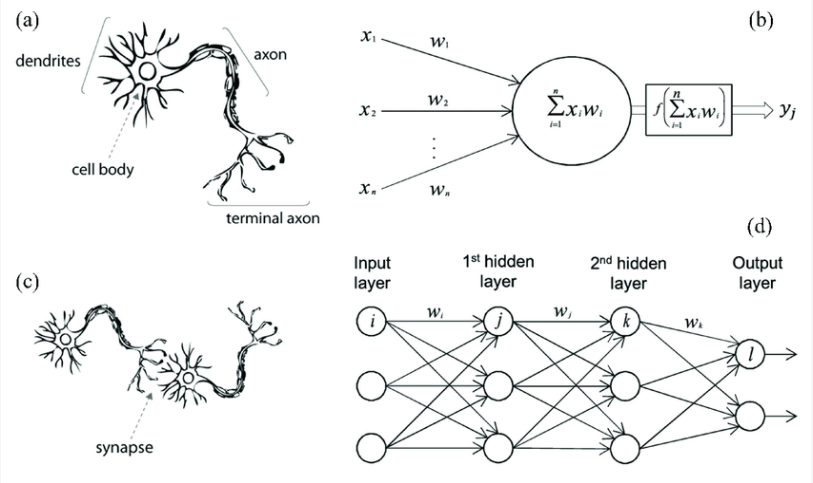
A neural network is a computational model inspired by biological neural networks in the human brain process information.

Purpose:

Used for tasks like classification, regression, and pattern recognition.

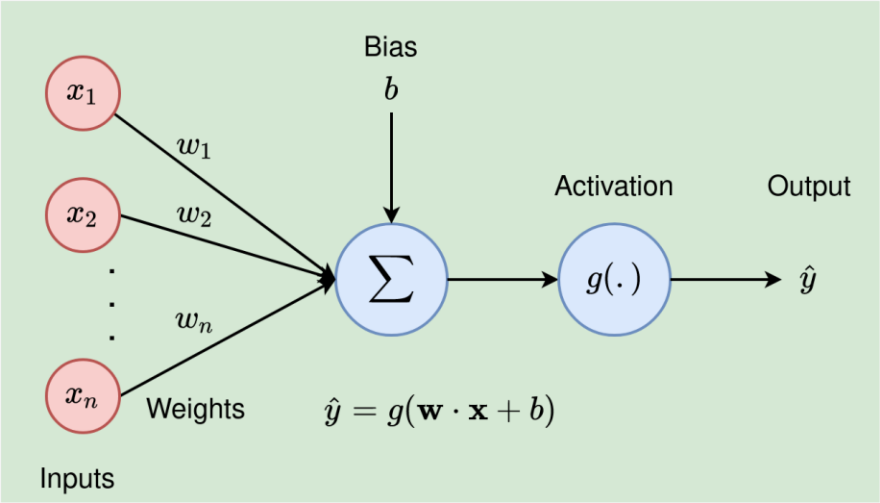
Components:

Composed of interconnected nodes (neurons) organized in layers.



Structure of a Neuron

- **Input:** Receives signals from other neurons or input data.
- **Weights:** Each input is multiplied by a weight, representing the strength of the connection.
- **Bias:** An additional parameter added to the weighted sum to adjust the output.
- **Activation Function:** Applies a non-linear transformation to the input.
- **Output:** The result is passed to the next layer or as the final output.



Activation Functions

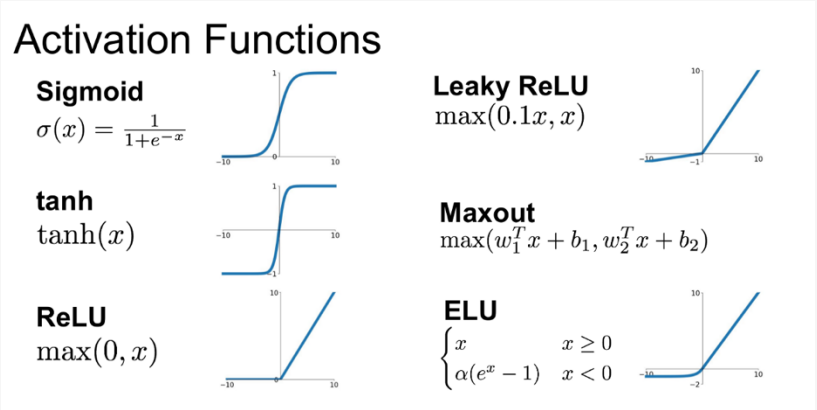
Purpose: Introduce non-linearity to allow the network to learn complex patterns.

Types:

- **Sigmoid:** Compresses values between 0 and 1.
- **Tanh:** Compresses values between -1 and 1.
- **ReLU:** Outputs input if positive, otherwise zero.
    - ReLU speeds up training by mitigating the vanishing gradient problem.

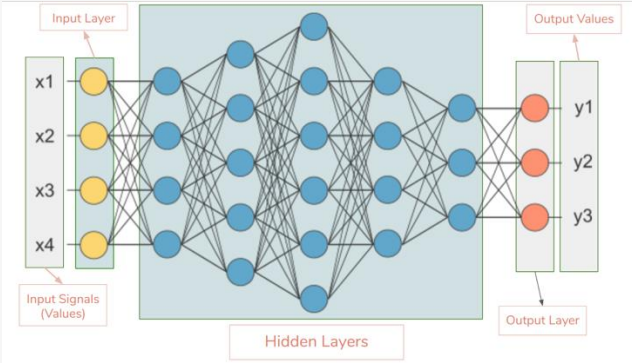
Choosing the right activation function can improve model accuracy.

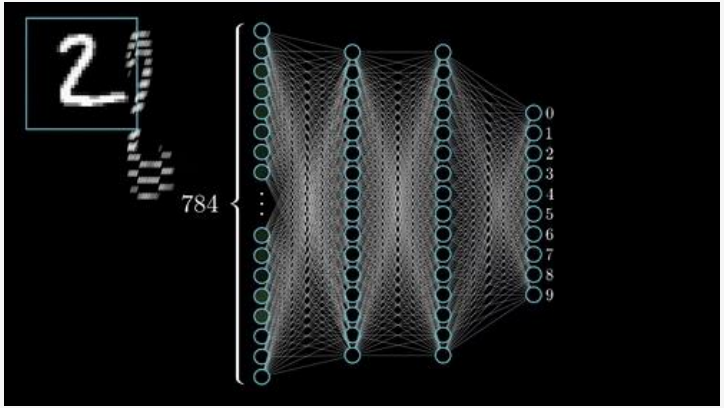
Visualization: Desmos



Layers in a Neural Network

- **Input Layer:** First layer that receives the input data.
- **Hidden Layers:** Intermediate layers that perform complex transformations.
- **Output Layer:** Final layer that provides the prediction or decision.





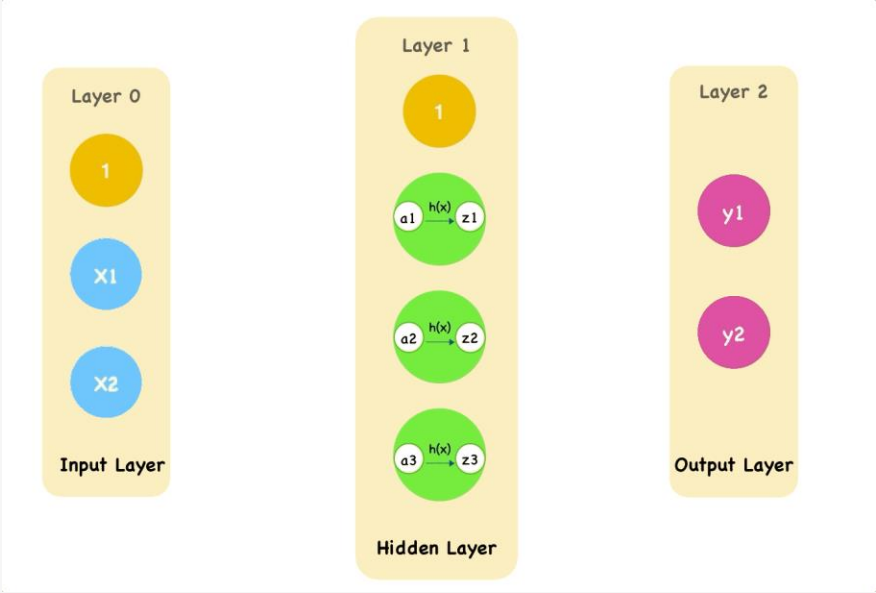
Forward Propagation Process

Definition:

Process by which input data is passed through the network layers to produce an output.

Steps:

- **Step 1:** Input features are fed into the input layer.
- **Step 2:** Weighted sums are calculated in each neuron.
- **Step 3:** Activation functions are applied.
- **Step 4:** The final output is generated in the output layer.



Backpropagation and Gradient Descent

Backpropagation:

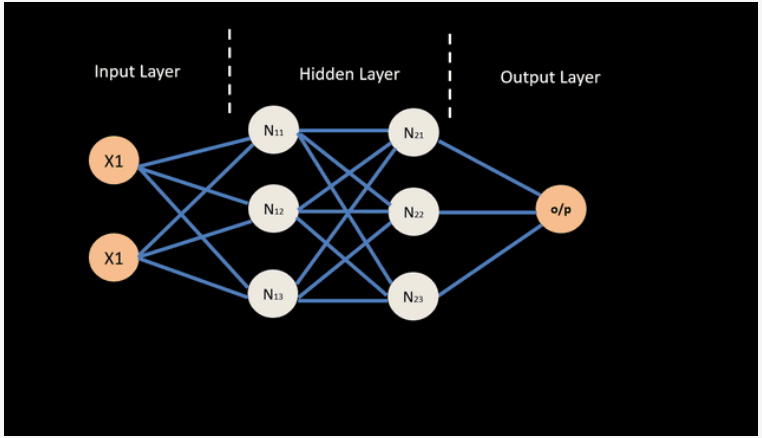
Technique to adjust the weights of the network based on the error calculated by the loss function.

Gradient Descent:

Optimization method used to minimize the loss function by iteratively adjusting the weights.

Steps:

- **Step 1:** Calculate gradients.
- **Step 2:** Update weights.
- **Step 3:** Repeat until convergence.



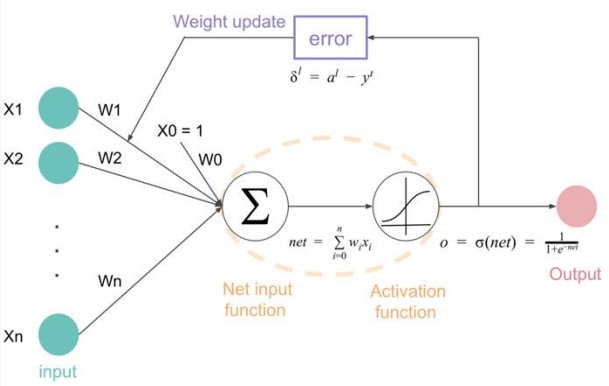
Loss Function

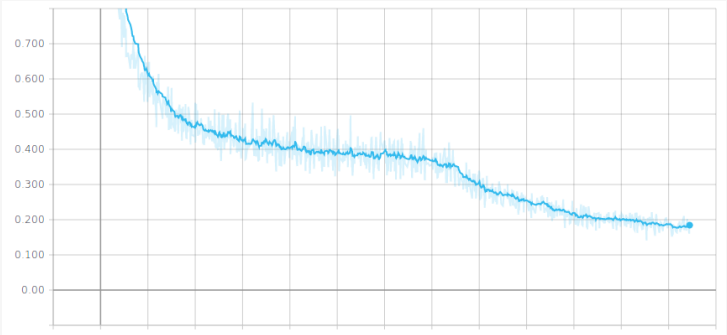
Definition:

A loss function measures how far the network's predictions are from the actual target values.

Types:

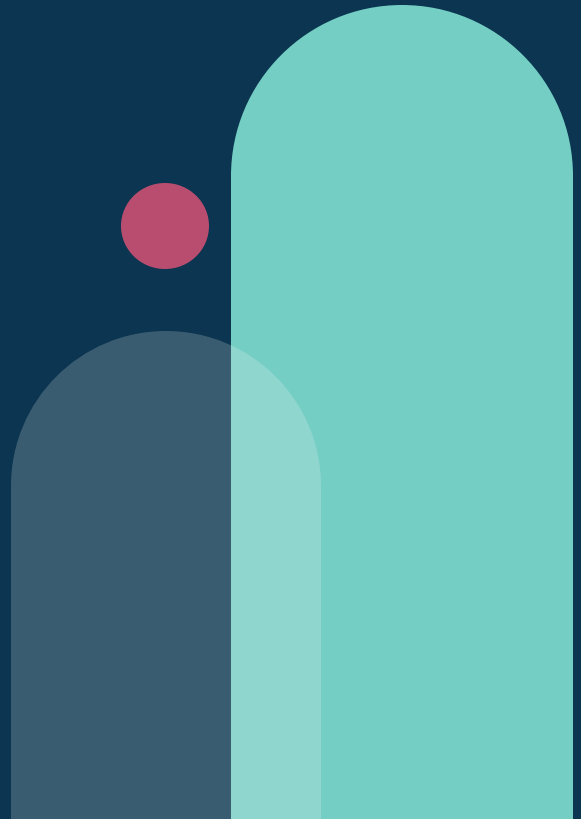
- **Mean Squared Error (MSE):** Used for regression tasks.
- **Cross-Entropy Loss:** Used for classification tasks.
- **Purpose:** Guides the training process by providing feedback on prediction accuracy.
- **Optimization Goal:** Minimize the loss function to improve model predictions.





02

ML/DL Algorithms

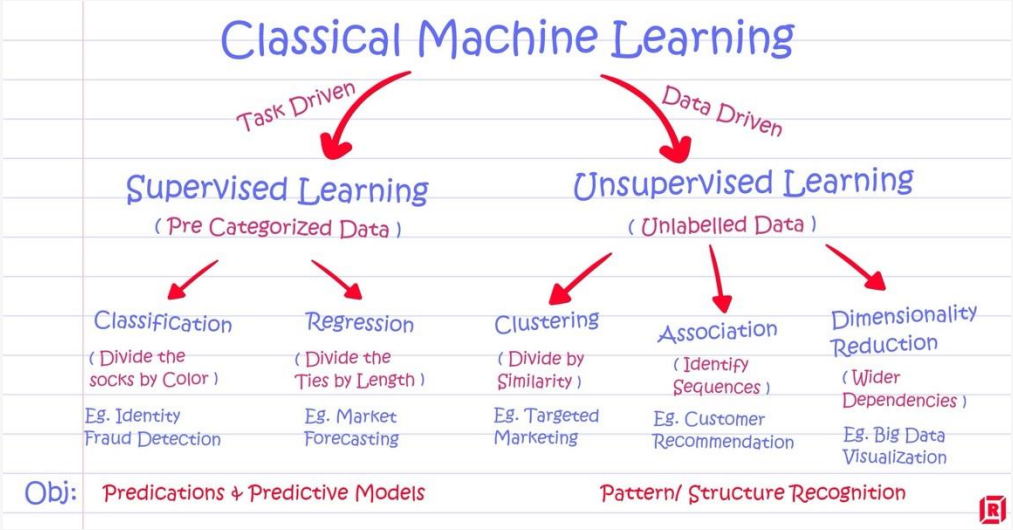


Overview of Key ML Algorithms

What are ML Algorithms?
Algorithms that enable machines to learn from data and make predictions or decisions without being explicitly programmed for the task.

Supervised Learning:
Algorithms that learn from labelled data.

Unsupervised Learning:
Algorithms that identify patterns or groupings in unlabelled data.



Introduction to Deep Learning Algorithms

Deep Learning:

A subset of machine learning that uses neural networks with multiple layers (deep architectures) to model complex patterns in data.

Why Deep Learning?

Capable of automatically learning feature representations from raw data, eliminating the need for manual feature extraction.

Popular DL Networks:

- **Convolutional Neural Networks (CNNs):** Specialized for image processing.
- **Recurrent Neural Networks (RNNs):** Designed for sequence data, such as text or time series.

Types of Neural Network Architectures

Feedforward Neural Networks (FNN):

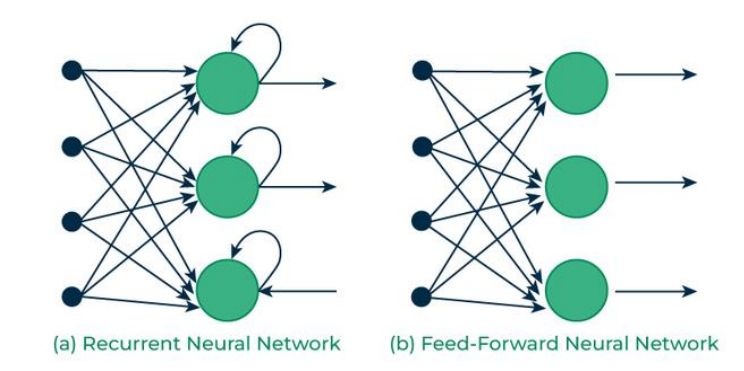
- Data flows in one direction from input to output.
- Commonly used for tasks like image recognition.

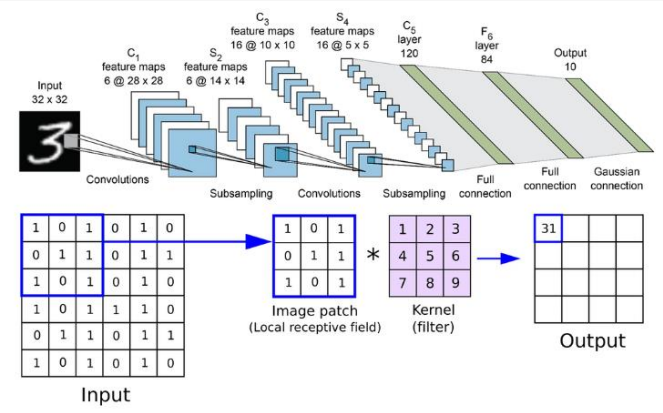
Recurrent Neural Networks (RNN):

- Includes loops allowing information to persist, useful for sequential data like time series.

Convolutional Neural Networks (CNN):

- Specialized for grid-like data such as images.





Convolutional Neural Networks (CNNs)

Purpose:

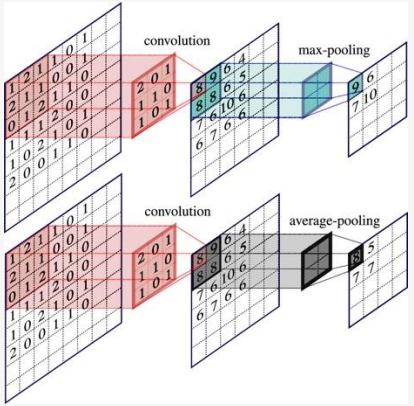
CNNs are primarily used for tasks that involve grid-like data, such as images.

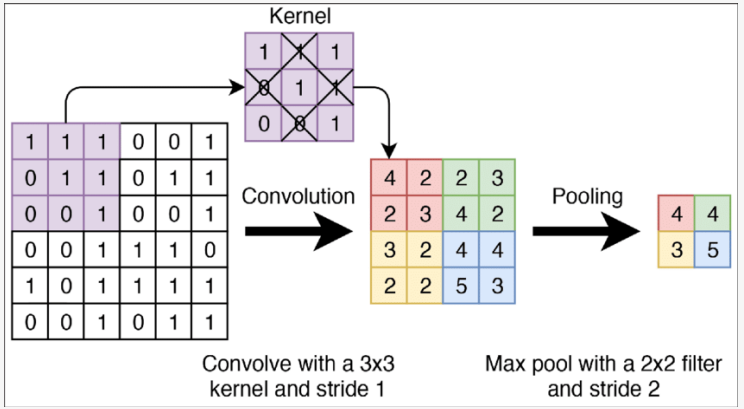
Key Components:

- **Convolutional Layers:** Detect patterns like edges and textures.
- **Pooling Layers:** Reduce the dimensionality, preserving important features.

Use Case:

- Image recognition and classification, e.g., identifying objects in a photograph.





Recurrent Neural Networks (RNNs)

Purpose:

RNNs are designed for sequence data, where the order of the data points matters.

Key Feature:

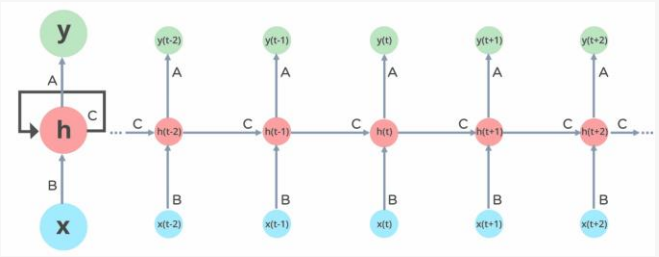
Recurrent Connections: Allow information to persist, enabling the model to remember previous inputs.

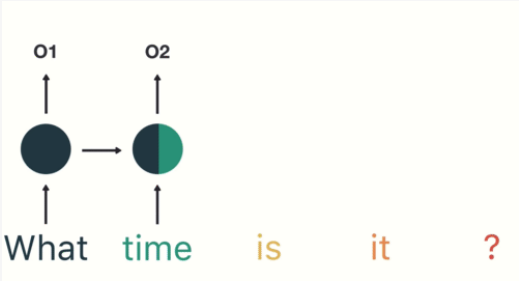
Variants:

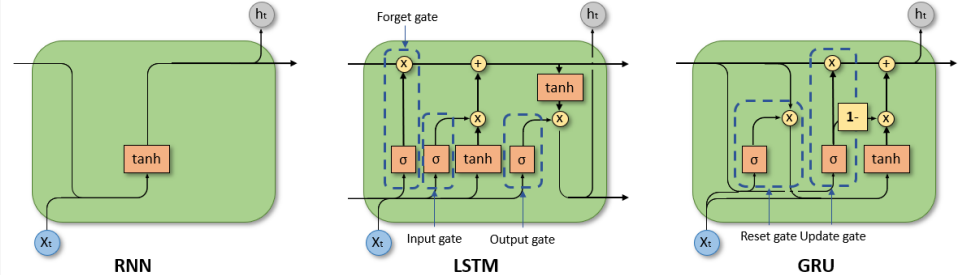
- **LSTM (Long Short-Term Memory):** Helps with long-term dependencies in sequence data.
- **GRU (Gated Recurrent Units):** A simplified version of LSTM, often faster to train.

Use Case:

Language modeling, such as predicting the next word in a sentence or generating text.

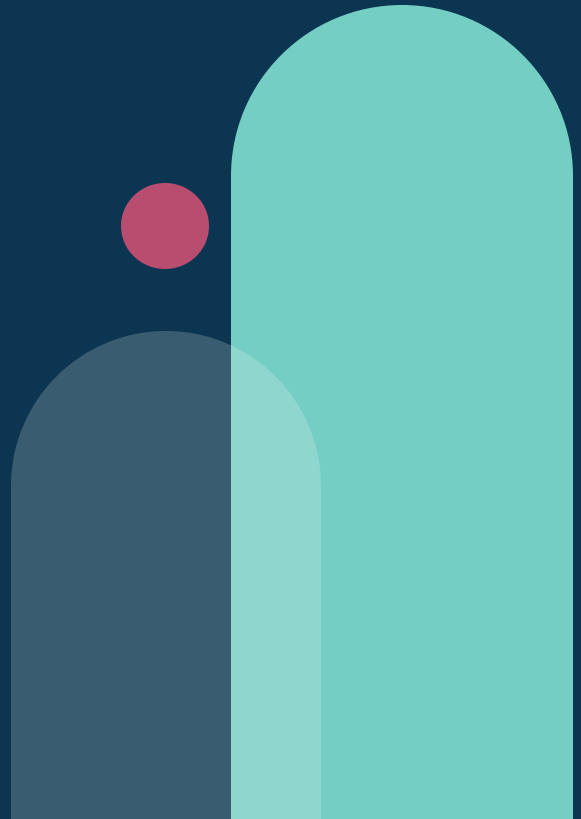






03

Vectors and Embeddings



Introduction to Vectors

What are Vectors?

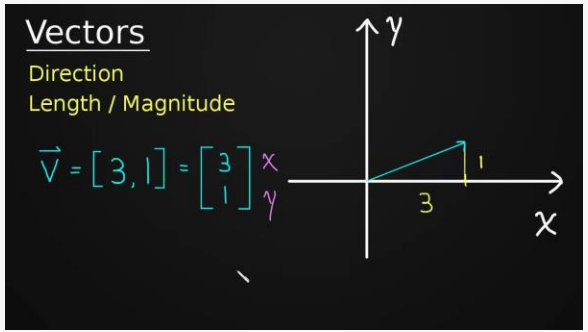
Vectors are mathematical representations of data points in a continuous space, often used to represent features or attributes in machine learning.

Why Vectors?

Vectors enable mathematical operations like addition, scaling, and dot products, which are fundamental to many machine learning algorithms.

Applications:

Used in tasks like document similarity, image processing, and recommendation systems.



https://www.desmos.com/calculator/u8wt5rnw9n
https://www.math3d.org/obnDo2J1w

Embeddings

What are Embeddings?

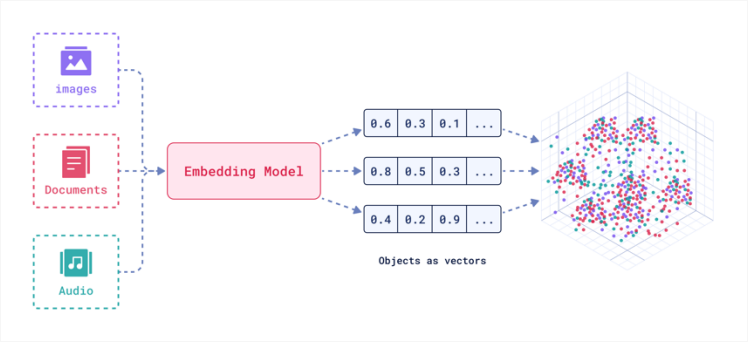
Embeddings are dense vector representations of data, particularly high-dimensional categorical data like words.

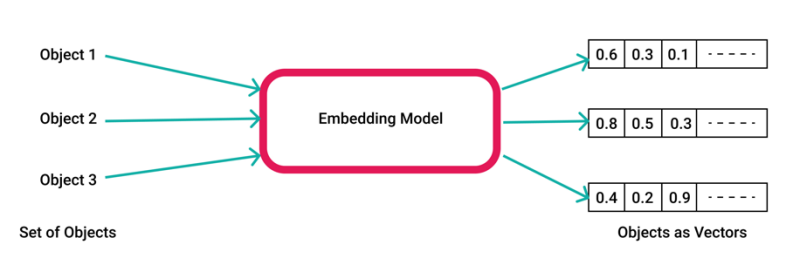
Purpose:

Reduce the dimensionality of data while preserving its semantic relationships.

Types of Embeddings:

Word Embeddings: Represent words in a continuous vector space, capturing their meanings.





Dimensionality Reduction

What is Dimensionality Reduction?

The process of reducing the number of input variables or features in a dataset while retaining the most important information.

Why is it Important?

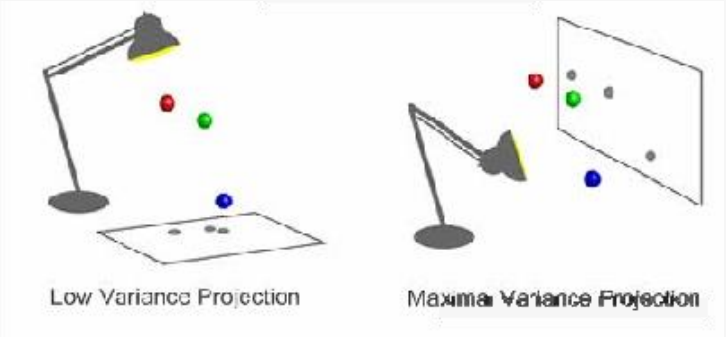
Simplifies models, reduces computational cost, and helps in visualizing high-dimensional data.

Key Techniques:

- **Principal Component Analysis (PCA):** Projects data into a lower-dimensional space by identifying the directions (principal components) that capture the most variance.
- **t-SNE (t-Distributed Stochastic Neighbor Embedding):** A technique used for visualizing high-dimensional data by reducing it to two or three dimensions.

https://projector.tensorflow.org/





Word Embeddings

Word Embeddings in NLP:

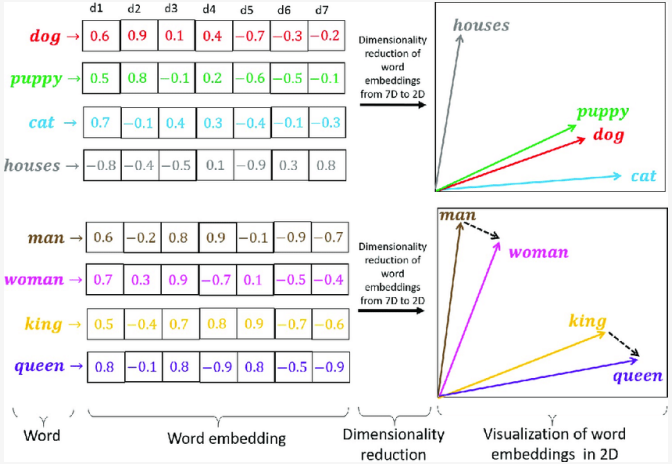
Transform words into vectors that capture their meanings, context, and relationships to other words.

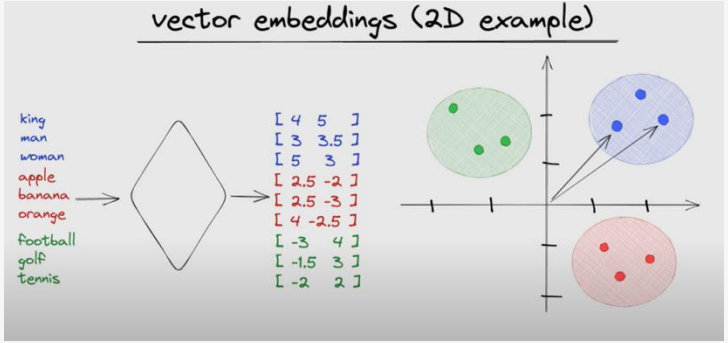
Key Techniques:

- **Word2Vec:** Predicts a word based on its context or vice versa.
- **GloVe:** Generates word vectors by analyzing word co-occurrence in a large corpus.
- **FastText:** Extends Word2Vec by considering subword information, helping with rare words and misspellings.

Importance in NLP:

Enable models to perform better in tasks like sentiment analysis, translation, and question answering.





Recap and Key Takeaways

Recap:

- Introduced vectors as fundamental representations in ML.
- Explained embeddings and their importance in capturing relationships within data.
- Discussed dimensionality reduction techniques like PCA and t-SNE.
- Covered practical examples of word embeddings and dimensionality reduction.

Key Takeaways:

- Vectors and embeddings are crucial for representing complex data in a way that machine learning models can understand.
- Dimensionality reduction techniques help simplify data and make it more interpretable.

Next Steps:

- Prepare for the next session, where we'll dive deeper into practical applications and introduce advanced topics like vector search.

Thanks!

Do you have any questions?

